

Eli Boyden

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EDUCATION

University of Virginia

B.S. Computer Science & B.A. Mathematics, Data Science Minor | GPA: 3.95/4

Charlottesville, VA

Expected Graduation: May 2027

EXPERIENCE

Software Developer Intern

May 2026 – August 2026

SAS Institute

Cary, NC

- Designed and implemented unit tests in Golang, increasing SonarQube test coverage on production services from 54% to 88% and identifying multiple defects leading to fixes on production code.
- Diagnosed and fixed GitHub Actions failures blocking CI/CD on main branches, restoring reliable automated builds for production services.

Software Engineer Intern

August 2025 – December 2025

City of Charlottesville Department of IT

Charlottesville, VA

- Integrated PostgreSQL cloud database with Supabase, creating a RESTful API for storage and manipulation of alert updates and user authentication data.
- Accelerated cloud adoption by automating collection, processing and analysis of API and RSS feed data on critical events in Charlottesville using specialized Python AWS Lambda serverless functions.

Full Stack Software Engineer Intern

May 2025 – August 2025

YSTEM and Chess

Remote

- Wrote and maintained automated unit and integration tests using Jest, reducing React front-end production bugs and increasing development team test coverage by over 60%.
- Led accessibility improvements across React components, increasing compliance with WCAG standards, and improving usability for screen readers, keyboard navigation, and users with visual impairments.

Application Software Developer

May 2025

Mathesis Technology

Hybrid Charlottesville, VA

- Customized and deployed a Markdown-to-HTML static site generator to Cloudflare, launching mathematics course websites for students and instructors, and increasing digital math accessibility with MathML.
- Coauthored and edited a peer-reviewed mathematics education [paper](#) published on arXiv and in the American Society for Engineering Education (ASEE) Journal.

PROJECTS

TAGuide | PostgreSQL, Python, Encryption (bcrypt, Fernet), Flask, React

- Full-stack React application for running Selenium web scraping jobs to automate data collection from Gradescope, enabling tracking and visualization of Teaching Assistant (TA) grading progress.
- Helped identify and subsequently correct grading work imbalances of up to 25%.

Distributed Routing Algorithms | Golang

- Implemented spanning-tree and distance-vector routing algorithms in Go, simulating distributed network nodes that exchange messages to elect roots, discover routes, and maintain forwarding tables.
- Built a discrete-event simulation using a generic weighted-graph abstraction and min-heap event queue to model asynchronous message propagation, route updates, and network latency.

SKILLS

Programming Languages: Python, Java, Go, JavaScript, TypeScript, SQL, C, HTML, CSS, SCSS (Sass)

Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Jest, JUnit, JPA, Selenium, Gradle, Django

Tools & Platforms: Git/Github, AWS Lambda, Docker, Kubernetes, Cloudflare, Github Copilot, Claude, Codex

Concepts & Methodologies: Object-Oriented Programming (OOP), Agile Development, Scrum, Unit Testing, Accessibility (WCAG), REST APIs, CI/CD, Machine Learning

Databases: PostgreSQL, MongoDB, SQLite, Supabase

CERTIFICATIONS

AWS: Certified Cloud Practitioner CLF-C02 | [Certification](#)

Scrimba: Advanced React | [Certification](#)